

Scaling Sensing with Physical Transport Units (PTUs)

Silicon for scientists and AI to observe the physical world at extreme scale

Almost every scientific instrument today follows the same block diagram:

Physical World → Sensors → [Data Transport] → Compute (often GPUs) → Intelligence

A spectrometer, microscope, gene sequencer, flow cytometer, radar, seismograph

The transport block, a digital fabric that buffers, formats, and transports raw sensor data into GPUs in real time, is where breakthroughs are stuck.

We are building **Physical Transport Units (PTUs)**, custom chips designed to scale the data transport layer between sensors and computation in frontier instruments. Targeting shared requirements like streaming, buffering, timestamping, and packetization at 0.1-1 TB/s lets one chip family serve hundreds of experiments. Unlike general-purpose chips that are currently used for data transport, we are focused on data movement and leave all computation to the destination GPU. This lets us produce a small, narrowly-scoped die with low power, cost, and complexity.

This is stuck. GPUs are the dominant realtime compute platform for science, but they can't receive real-world information directly. If you want to transport data from many elements, that role falls to reconfigurable logic chips that can lead to electronics being >80% of the cost of a large instrumentation project with many nodes, with individual nodes costing \$15-30k. Development is slow, and scientists spend years redeveloping pipelines from scratch for each new experiment.

As GPUs become more powerful, data ingress becomes relatively more expensive. However, ingress is exactly the capability scientists need to observe the physical world. Commercial chips optimize for consumer/business volume, not frontier science. In contrast, science benefits from orders of magnitude more real-time data, often with nonlinear returns to scale: phased arrays in radio telescopes, ultrasound in medical imaging, arrays of cheap accelerometers: all examples of scale and computation used to outperform a single, exquisite sensor.

Why now? Chiplet architectures let us build a chip from proven, separately-fabricated dies. To talk to GPUs, we can license the latest high-speed I/O chiplets designed and scaled for datacenter GPUs to talk to each other, and integrate them with configurable logic that adapts to a broad range of scientific sensor inputs. This dramatically reduces the tapeout risk and timeline, and upgrade cost.

AI-assisted chip design will compress cycles on the custom digital work. Together with chiplets, this lowers program costs from a \$100M+ from-scratch program to a \$20–30M focused effort over 18–36 months. The same PTU primitives of high-bandwidth ingress/egress, deterministic timing, and GPU packetization are all demanded by a broad range of sensing modalities like radar, ultrasound, radio telescopes, sonar, MRI, and brain-computer interfaces. But the cross-cutting opportunity here is too large for a single lab.

Although we believe PTU impacts will be largest in exploratory science – unlocking previously impossible instruments – we highlight three concrete examples.

Tornadoes. If you've ever received a tornado warning, it came from a mechanically-steered Doppler radar — decades-old technology that rotates physically and produces one volumetric scan every ~5 minutes. That buys ~13 minutes of average lead time, and it costs lives. Deep learning has transformed weather forecasting in just the last two years, with AI models now outperforming traditional numerical methods at the global scale — but tornado prediction needs something different: high-rate, high-fidelity local radar data, which essentially doesn't exist. The technology exists as research demos, but going digital has been too costly to deploy nationally, and roughly half the cost is the digital fabric between sensors and computing. It's low-volume, specialized hardware, so it stays expensive.

Cancer. Your typical clinical ultrasound scanner runs 64-128 channels through an onboard beamformer that collapses raw per-element signals into a single image stream, before they reach a computer. A 256-channel Verasonics research scanner that captures raw voltages from every element instead runs \$300K–\$1M. Streaming all of that data into a GPU unlocks capabilities for collecting rich datasets that beamformers throw away, like tissue microstructure for finding liver disease without a biopsy, super-resolution imaging that maps capillaries finer than a human hair, and real-time temperature inside tumors during ablation. With commodity sensors, the cost is dominated by the fabric connecting sensors to compute. It's low-volume, specialized hardware, so it stays expensive.

The Universe. My own field of astrophysics is now scaling from single exquisite instruments to arrays of 10^3 - 10^6 cheap antennas, digitally combined into a synthetic dish that looks in every direction at once. A five-year survey with this million-antenna telescope would 10x the observable volume of the Universe humanity has ever seen, directly test the nature of dark matter and dark energy, and discover thousands of exoplanets along the way. A panel rejected a similar concept a few years ago at \$600M; ninety percent of that cost was the fabric from sensors to GPUs. It's low-volume, specialized hardware, so it stays expensive.

Scaling instruments in this way leads to unusually high-conviction opportunities: the fundamental science usually isn't the risk. But it's impossible for a weather or ultrasound lab to create the missing chip to make their work economically scalable at the conventional \$100M+ pricetag.

Risks. The primary technical risk is verification and integration. Advanced-node chips are tricky, but the playbook is well-known and chiplet architectures with licensed IP substantially reduce the first-tapeout risk. We'll iterate with multi-project wafer runs and partner with industrial semiconductor vendors for mixed-signal IP and advanced packaging. We'll lower the adoption risk with co-design and in-kind support, shipping development boards, drivers, GPU-side software, and reference circuit boards so labs can focus on science.

Cost and Timeline. Design and first tape-out will cost \$20–30M, depending on IP reuse and eventual ownership: roughly \$5–8M for IP licensing, \$6–10M for mask sets, packaging, and MPW

runs, and \$8–12M for engineering across digital design, verification, and software. This includes production development boards as well as GPU-side driver and software development. Most scientific labs lack high-speed PCB expertise, so this scaffolding is essential for translational impact.

Phase 1 (1-12 months) is digital design validation on commercial reconfigurable chips, as well as lab testing of commercially-available IO chiplets. We would build an MVP at 10-50x unit cost with the essential capabilities of data ingress/egress, and begin co-design with instrument-building labs across domains. We plan to seed several exemplar projects, offering in-kind support (chips plus integration) and partner for external instrument funding.

Phase 2 (12–18 months) is silicon design, verification, first tape-out on a 12nm process, and chiplet interposer integration. Deliverables are working silicon on development boards plus reference PCBs — the primary interface for experimental labs — along with drivers, documentation, and GPU-side software co-developed with GPU vendors.

Phase 3 (18–48 months) is pilot instrument integrations with partner labs across disciplines — testing drivers, documentation, and GPU-side software, and validation in real experimental environments. This would involve broad deployment across fields, a second-gen chip incorporating lessons from pilot integrations, and an open-source software ecosystem.

The real measure of our success is science impact: impossible experiments brought to life, surprising insights, and lives saved. Our month 3 MVP is a working prototype on commercial reconfigurable development chips, streaming 64 GB/s of sensor data to a GPU with real-time processing. I've worked on two experiments that built smaller-scale versions of this pipeline! The midterm at month 18 is to build working silicon that streams 100 GB/s into a real instrument, with scientists in pilot labs exploring development board usage.

The final exam is a world of scalable instruments, built for collecting the datasets we need to understand the physical world with tools like deep learning. Powerful arrays for understanding storms, research-grade ultrasound in every rural hospital, and telescopes opening up the mysteries of dark matter and dark energy.

Zack Li, astrophysicist at UC Berkeley (zackli@berkeley.edu)

*Developed through the **Renaissance Philanthropy** Big if True Science Accelerator*